In silico protein evolution with reinforcement learning

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I. Challenge

Protein fitness landscapes are high-dimensional, discrete, and rugged: single amino acid substitutions can dramatically alter stability and activity, yet exhaustive experimental characterization is intractable. Deep mutational scanning (DMS) provides high-quality functional measurements but covers only a vanishingly small fraction of sequence space. For example, a published adeno-associated virus (AAV2) DMS experiment targeting 28 residues defines a combinatorial space of $20^{28} \approx 2.7 \times 10^{36}$ variants, while only 42,328 mutants were experimentally assayed [2].

Foundation protein language models (PLMs), such as ESM-2 [6], embed protein sequences into high-dimensional latent spaces that capture evolutionary and structural constraints. While these representations correlate well with biophysical properties such as stability and foldability, they are not optimized to predict protein-specific functional activity [4]. Prior work has therefore relied on hybrid computation–experimentation loops: random or heuristically sampled variants are embedded with PLMs, ranked by supervised regressors (e.g., random forests), assayed experimentally, and iteratively used to retrain the predictor [5]. However, these approaches rely on dense experimental feedback and do not explicitly learn how to navigate the combinatorial structure of sequence space.

This motivates the need for search algorithms capable of efficient exploration under sparse, delayed rewards. Reinforcement learning (RL) provides an apt framework: protein sequences define states, mutations define actions, and experimental stability and activity provide reward. Unlike purely supervised pipelines, RL policies can learn search strategies that balance local exploitation with global exploration, enabling principled navigation of vast fitness landscapes [7].

II. Problem

We investigate whether an RL agent can learn an effective mutation policy for optimizing an AAV2 capsid protein given limited experimental data, namely a DMS dataset. AAV2 capsids are highly desirable targets for engineering as they are used in FDA-approved therapeutics and investigational drugs as the delivery vehicle for various therapeutic and vaccine payloads. Increasing viral production efficiency through intentional mutation can ultimately lower costs and reduce the required dosage for patients by boosting yields and improving the quality of AAV2 therapeutics.

We begin with a published DMS dataset that mutates a 28-amino-acid subsequence of the 735-amino-acid AAV2 capsid protein. This subsequence is known to span buried, surface, and interface regions and overlaps both known heparin- and antibody-binding sites, and thus serves as an interesting target for mutation from a biological standpoint, while also reducing computational complexity [2].

An agent iteratively edits a wild-type sequence to maximize a composite reward r_t combining:

(i) experimental DMS activity, when available; otherwise, we query a surrogate model trained on DMS to predict out-of-DMS activity from ESM-2 embeddings

(ii) ESM-2 log likelihood of proposed variant compared to wild-type.

Formally, let $x \in \mathcal{A}^{28}$ be a sequence of length 28 and $\mathcal{A}$ the alphabet of 20 canonical amino acids. At each time step t, the agent observes state s_t and selects a mutation

$$a_t = (i, a) \quad i \in \{1, \ldots, 28\},\ a \in \mathcal{A},$$

producing s_{t+1}. Episodes terminate after 6 time steps. The objective is to maximize the expected reward

$$J(\pi) = \mathbb{E}_\pi \Big[\sum_{t=1}^{6} \gamma^{t-1} r_t \Big].$$

III. Mathematical Formulation

A. State and Action Spaces

- **State** $s_t \in \mathcal{A}^{28}$: the current sequence, represented as a length-28 vector of indices.

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- **Action** $a_t = (i, a)$: a point mutation at residue i substituting amino acid a. The action space is the Cartesian product $\{1, \ldots, L\} \times \{1, \ldots, 20\}$, formally represented as a MultiDiscrete space in Gymnasium.

B. Rewards

The reward associated with a variant x at any timestep is calculated as

$$r(x) = \frac{1}{2} \cdot \frac{\text{fitness}(x) - \bar{\mu}_{\text{fitness}}}{\bar{\sigma}_{\text{fitness}}} + \frac{1}{2} \cdot \frac{\text{activity}(x) - \mu_{\text{activity}}}{\sigma_{\text{activity}}}$$

1. Fitness definition

To quantify the biological plausibility of a proposed mutant sequence relative to wild-type, we compute a masked pseudo-log-likelihood (PLL) score using pre-trained ESM-2 (8M). Let $x^{\text{wt}}, x^{\text{mut}} \in \mathcal{A}^{735}$ denote the wild-type and mutant sequences of full AAV2 length 735. Define the set of mutated positions as

$$M = \{i \in \{1, \ldots, 735\} \mid x_i^{\text{wt}} \neq x_i^{\text{mut}}\}.$$

For each $i \in M$, we construct masked versions of both sequences by replacing position i with a mask token. ESM produces a vector of logits over amino acids at the masked position, defining a categorical distribution

$$p(a \mid x_{\setminus i}) = \text{softmax}(\ell(x^{\text{mask}(i)}))_a,$$

where $\ell(\cdot)$ denotes the extracted logits and $x^{\text{mask}(i)}$ is the sequence with position i masked. We extract the log-probabilities assigned to the true residues at position i in both sequences and compute their difference:

$$\Delta_i = \log p\left(x_i^{\text{mut}} \mid x_{\setminus i}^{\text{mut}}\right) - \log p\left(x_i^{\text{wt}} \mid x_{\setminus i}^{\text{wt}}\right).$$

The PLL ratio between the mutant and wild-type sequences is then

$$\text{PLL}(x^{\text{mut}}, x^{\text{wt}}) = \sum_{i \in M} \Delta_i.$$

This formulation yields a fast, local approximation of the relative likelihood of the mutant without requiring full-sequence likelihood computation.

$\bar{\mu}_{\text{fitness}}$ and $\bar{\sigma}_{\text{fitness}}$ are calculated as the sample mean and sample standard deviation, respectively, of the PLL of 100 randomly sampled sequences from our DMS dataset.

2. Activity definition

$$\text{activity}(x) = \begin{cases} \text{DMS}(x) & \text{if } x \text{ observed in DMS dataset} \\ \widehat{\text{DMS}}_\theta(x) & \text{otherwise} \end{cases}$$

where $\widehat{\text{DMS}}_\theta$ is a multi-layer perceptron whose architecture is explained in Section IV. *B.* μ_{activity} and σ_{activity} is calculated as the mean and standard deviation, respectively, of all scores in the DMS dataset.

C. Episode Dynamics

Episodes begin at the wild-type s_0 and proceed for 6 mutations. This choice stems from the observation that over 90% of proteins in the published AAV2 DMS dataset contain ≤ 6 mutations from wild-type (Fig. I). Additionally, prior modeling has shown close to 100% AAV2 viability at 6 mutations from wild-type, a threshold beyond which variants are largely nonviable [2].

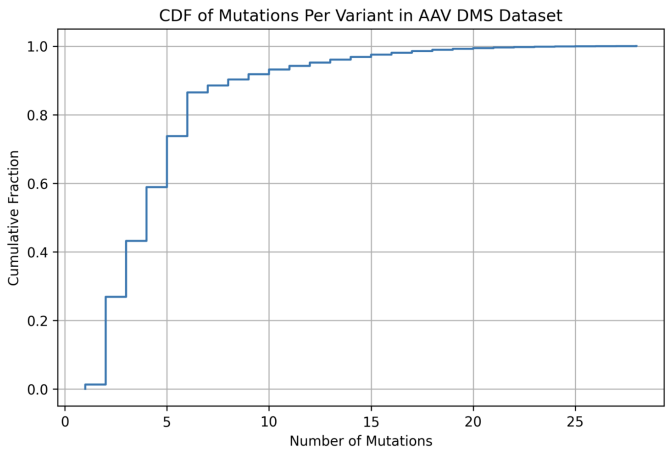

FIGURE I: CUMULATIVE DISTRIBUTION FUNCTION FOR NUMBER OF MUTATIONS PER PROTEIN IN AAV2 DMS DATASET.

This may indicate that further mutation strays too far from wild-type to function productively, or this may simply be an artifact of the high-throughput wet lab mutation process; in the first case, it is unproductive to stray beyond 6 mutations, and in the second case, it is untrustworthy to train a surrogate model on the impoverished datapoints of proteins with > 6 mutations.

Once the number of steps exceeds 6, we set a truncation condition to `True`. We use truncated trajectories, rather than absorbing terminal states, because it is incorrect to claim that a given protein is "done mutating" after 6 steps. Instead, it is appropriate to artificially end mutation after 6 steps due to aforementioned coverage of the majority of productive mutations.

IV. METHODS

A. Environment Design

We implemented a custom `ProteinEnv` in `stable-baselines3` supporting:

- multi-environment parallelism
- tuple-encoded action decoding (i, a)

- reward origin tracking (DMS versus surrogate)
- mutation history logging for interpretability

ESM-2 embeddings were precomputed for all positions of the wild-type and updated using single-position substitutions, enabling significant compute savings relative to recomputing full transformer passes.

B. *Surrogate Fitness Model*

We trained a regressor, $\widehat{\text{DMS}}_\theta$, on the AAV2 DMS dataset, with final $R^2 = 0.86$ after 10 epochs.

- Input: ESM-2 (8M) sequence embeddings, averaged across residues
- Architecture: 3-layer MLP ($320 \to 320, 320 \to 128, 128 \to 1$) with ReLU between each layer
- Loss: mean squared error on DMS scores

$$\mathcal{L}(\theta) = |\widehat{\text{DMS}}_\theta(x) - \text{DMS}(x)|^2$$

- Train/test split: 80/20

C. *Reinforcement Learning Algorithms*

We benchmarked four algorithms:

- PPO with $\lambda_{\text{GAE}} = 0.95$
- A2C with $\lambda_{\text{GAE}} \in \{0.0, 0.5, 1\}$.

All were run with $n_{\text{envs}} = 64, n_{\text{steps}} = 6, \gamma = 0.99$, and `total_timesteps` $= 10000$. Episodes were truncated at 6 mutations to match our biologically informed mutation budget.

V. Results

A. *Reward spikes periodically under imperfect truncation–rollout synchronization*

We initially observed periodic peaks in PPO and A2C performance every 7 rollouts (at rollouts 6, 13, and 20) (Fig. II).

Investigation revealed this to be a synchronization artifact of our training setup. Episodes were truncated when `step_count` > 6, while PPO collected rollouts of fixed length `n_steps` $= 6$. Every 7th rollout, the truncation boundary aligned with the rollout boundary, causing all environments to terminate simultaneously. We hypothesize that bootstrapping was more consistent at these points, producing trajectories with lower variance in advantage estimates. This likely made the policy update at those rollouts more stable, resulting in transient reward spikes.

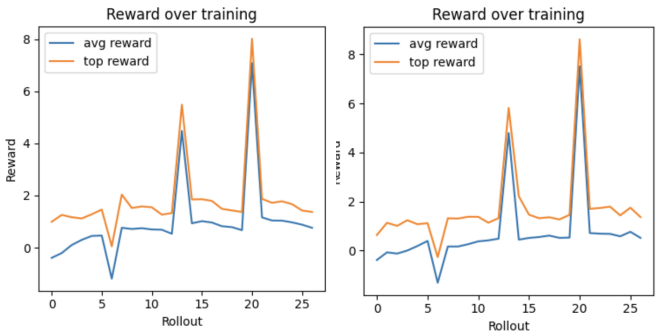

Figure II: Training reward with out-of-sync condition `step_count` > 6; A2C TD(1) on left, PPO on right.

B. *Models learn smoothly after truncation–rollout synchronization*

To test our hypothesis regarding synchronization, we changed our truncation condition to `step_count > 5`. After retraining all models across 10000 timesteps, reward spiking disappeared, and we observed smooth increase in reward over training. Further characterization of training metrics and learned policies revealed no other significant differences between the two conditions, confirming reward spiking was a unique artifact.

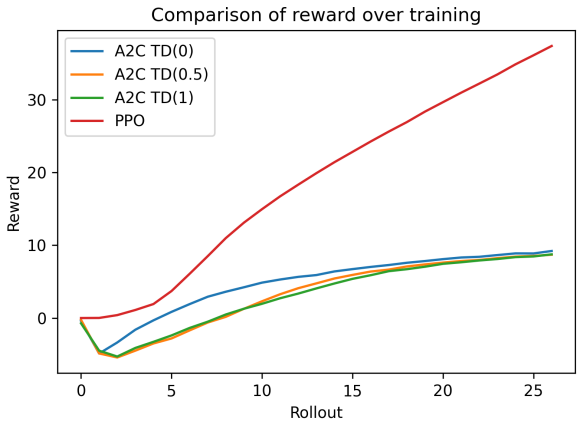

Figure III: Average reward curves for each algorithm.

In Figure III, PPO achieves nearly 3-fold higher reward at the end of training compared to all A2C variations. Interestingly, all A2C variations converge to similar maximum reward at the end of rollout 26. This behavior validates that our model trained on a positive trajectory, and that we have not reached saturation of our reward during training nor risked overfitting on our training data.

C. *Biologically grounded patterns emerge in mutational landscapes*

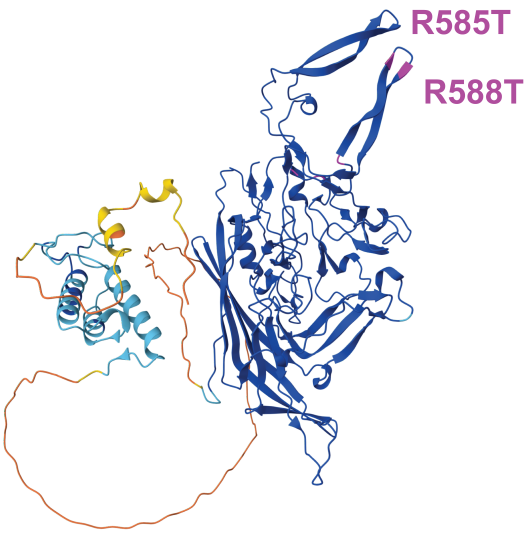

FIGURE IV: THE VARIANT WITH HIGHEST REWARD (2.1695) OUTPUT BY THE TRAINED PPO POLICY. ALL 5 POINT MUTATIONS ARE HIGHLIGHTED IN PINK, WITH HYPERVARIABLE REGION VIII MUTATIONS R585T AND R588T EMPHASIZED.

Across all algorithms, we observed consistent, non-random structure in the agent's mutation preferences. Positions 585 and 588 were mutated at substantially higher frequencies than all other sites, suggesting that the learned policies converged on specific, potentially biologically meaningful regions of the capsid.

To contextualize these residues, we visualized the AAV2 structure using AlphaFold3 (Fig. IV). Both sites localize to hypervariable region VIII (VR-VIII), which forms the tip of a highly exposed surface spike on the capsid. This region is solvent-accessible and structurally flexible relative to the capsid core, making it a plausible hotspot for mutations that can modulate viral properties without globally destabilizing capsid architecture [1].

This pattern is supported by prior experimental work. Hypervariable regions of AAV capsids are known to tolerate sequence variation across serotypes and have been widely used as engineering sites to alter viral properties while preserving assembly. In wild-type AAV2 specifically, R585 and R588 mediate binding to heparan sulfate proteoglycans (HSPGs), a primary cellular attachment factor. Substitutions at these positions disrupt HSPG binding and have been shown to detarget AAV2 from tissues such as liver and spleen [1]. Notably, these residues emerged as mutational hotspots in our experiments even though receptor binding or tissue tropism were not explicitly encoded in the reward function. This suggests that the learned activity–fitness landscape may implicitly reflect biologically meaningful structure–function constraints. It would also be of interest to experimentally evaluate our proposed variants in therapeutic gene delivery contexts to assess whether these computationally identified mutations translate to improved performance *in vivo*.

D. *A2C explores more novel variants than PPO during training, while PPO exploits actions with high reward*

Examination of heatmaps reflecting our action space revealed that all A2C models explored the action space significantly more exhaustively than PPO. In contrast, PPO honed in on certain actions nearly twice as frequently than any A2C model. We hypothesize that the observed difference arises from algorithmic factors. PPO's multi-epoch reuse of data and clipped surrogate objective tends to conservatively reinforce actions that perform well in the batch, which promotes early exploitation. In contrast, A2C's single-pass, multi-worker updates introduce higher variance and sustained stochasticity in the policy, producing broader exploration.

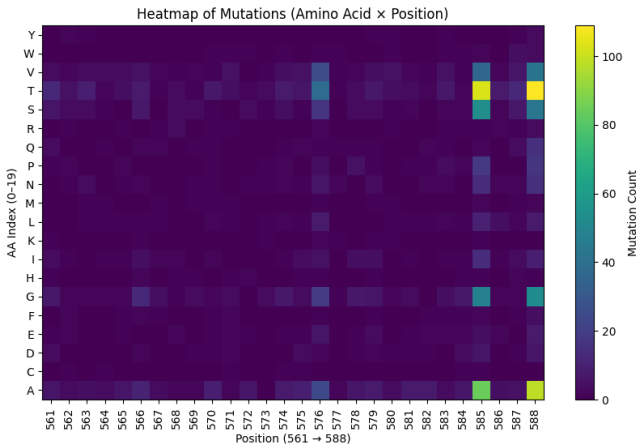

FIGURE V: PPO HEATMAP OF MUTATIONS OVER TRAINING.

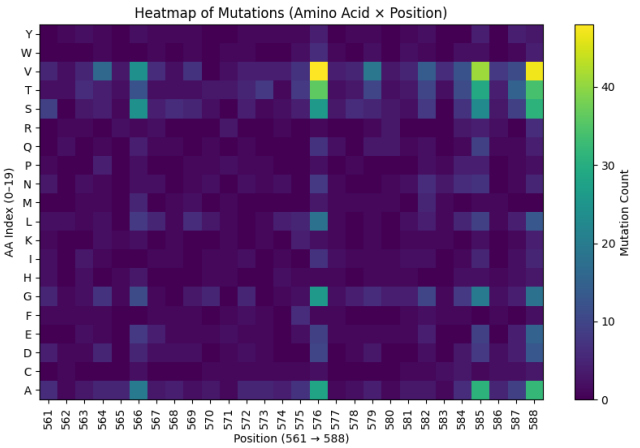

FIGURE VI: A2C TD(1) HEATMAP OF MUTATIONS OVER TRAINING.

To further quantify novelty, we defined a novelty score as the ratio of surrogate model calls to DMS queries during training. Indeed, during PPO training, the surrogate was queried 15 times more frequently than the DMS, whereas during A2C training, the surrogate was queried up to 344 times more frequently than the DMS (Table I).

E. *Generated variants lie in the top* 99.6–100th *percentile of reward distribution*

After training, we sampled 10 variants from each policy to characterize their reward in comparison to known variants in the DMS. This sample size was chosen to reflect a wet-lab–feasible validation regime, enabling future experimental validation. Mean reward per policy was calculated on these 10 variants (Table I). For each policy, we identified the variant with highest reward of the 10 sampled and calculated its percentile relative to 4000 randomly sampled DMS variants' rewards, as seen in Table I and Figure VII. We were pleased that our successful training led to variants with extremely high reward, meriting experimental validation.

	Novelty	Mean reward	Percentile
A2C TD(0)	107.00	1.3798	99.975
A2C TD(0.5)	89.95	1.3896	99.65
A2C TD(1)	344.60	1.3305	100
PPO	15.00	1.5101	100

Table I: Comparison between optimal model performance and heuristic baselines.

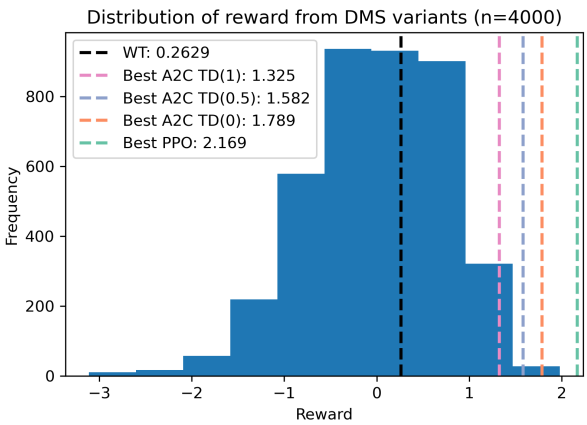

Figure VII: Distribution of the reward function sampled from 4000 sequences in the DMS dataset.

Despite sampling the action space the least thoroughly, PPO exhibits the highest mean reward of generated variants. This may indicate that compared to the A2C algorithms, PPO more effectively exploited mutations that it identified as beneficial early on.

VI. Future Steps

A key limitation of our approach is the surrogate model, which is imperfect and significantly impacts training. While we chose a simple MLP architecture and basic training strategy, future work could test more complex architectures to achieve a better performing surrogate [3].

We also note that our action space is not representative of all possible mutations. We only consider point mutations within a fixed 28-amino-acid section of the AAV2 capsid, known to be relevant for production efficiency and functionality; however, it may be more impactful to allow mutation across the entire sequence. Furthermore, we ignore the potentially beneficial effects of insertion and deletion mutations, instead favoring substitutions for simplicity; the MDP would need to be modified to accept expanded mutation types.

RL policies are black boxes, so it would be illuminating to pair our models with an LLM capable of explaining why certain mutations are favorable. Beyond probing the RL black box, this may elucidate fundamental principles of protein biology. In parallel, wet lab validation of our generated mutants and revealed principles would support the suitability of RL in this setting.

Finally, A2C and PPO are merely two classes of RL policy optimization algorithms. Comparison with other families of algorithms and benchmarking against non-RL algorithms may contextualize our preliminary results.

VII. Conclusion

This work explores whether modern RL algorithms can serve as efficient search strategies for protein engineering when supported by DMS and PLM-based activity modeling. By framing sequence design as a Markov decision process and combining DMS-grounded rewards with PLM-derived predictions, we construct an agent capable of exploring mutation space under limited supervision. Early results demonstrate coherent mutation patterns, recovery of biologically meaningful residue positions, and algorithm-dependent exploration dynamics.

VIII. Code Availability

The code for this project is available at `https://github.com/jeannieshe/protein-evolution`.

References

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IX. Appendix

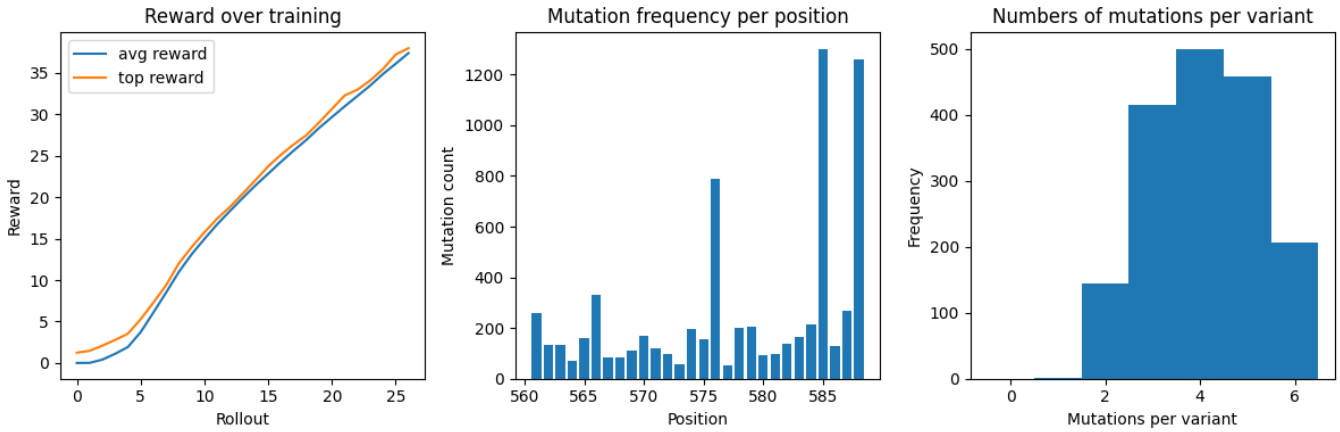

Figure VIII: PPO training metrics

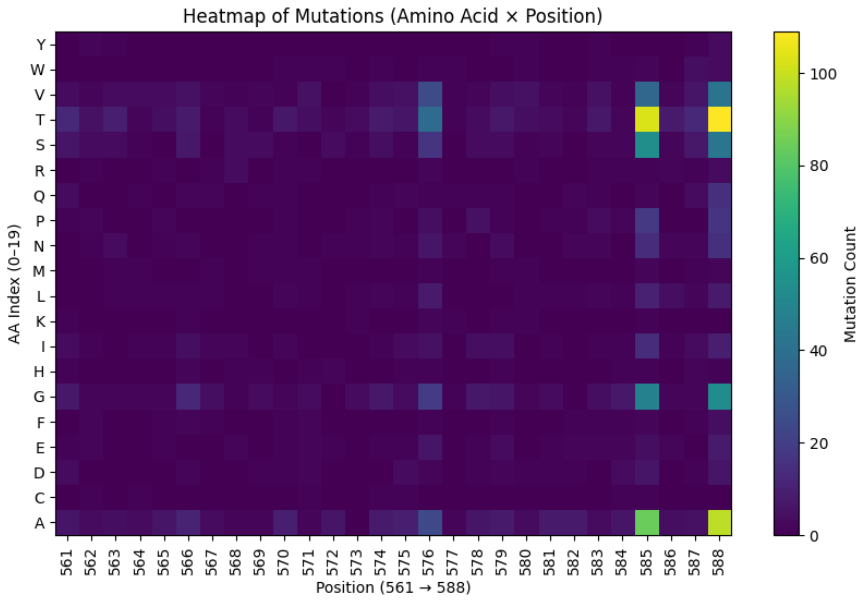

Figure IX: PPO heatmap of mutations over training

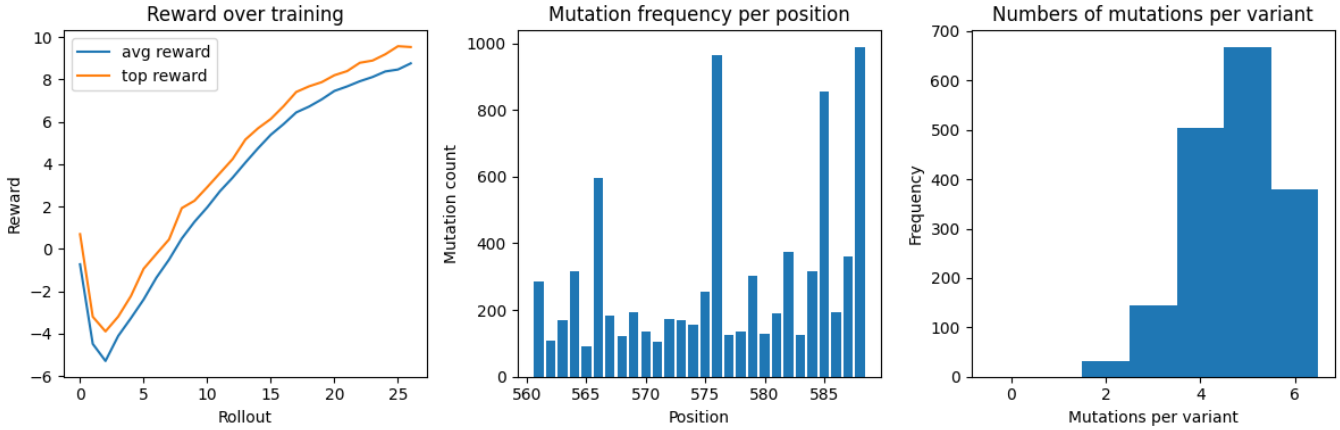

Figure X: A2C TD(1) training metrics

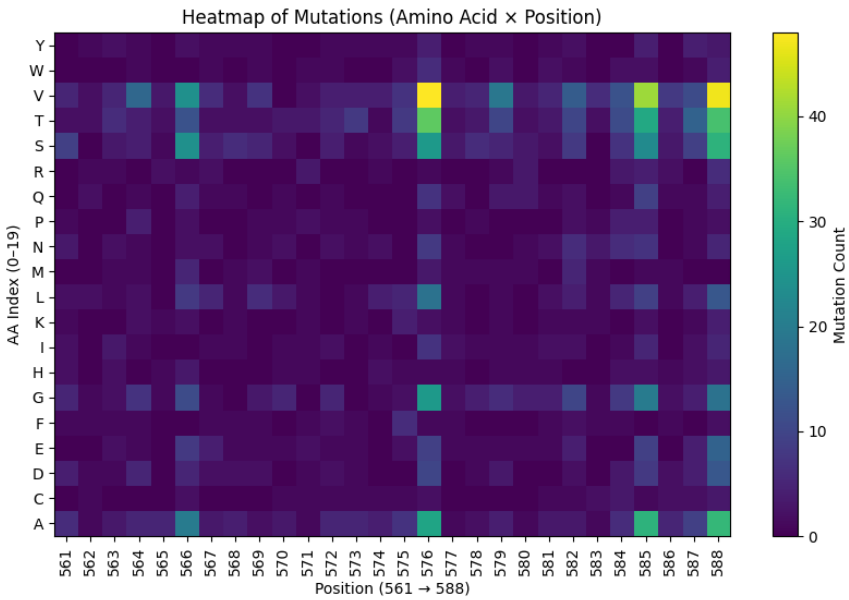

Figure XI: A2C TD(1) heatmap of mutations over training

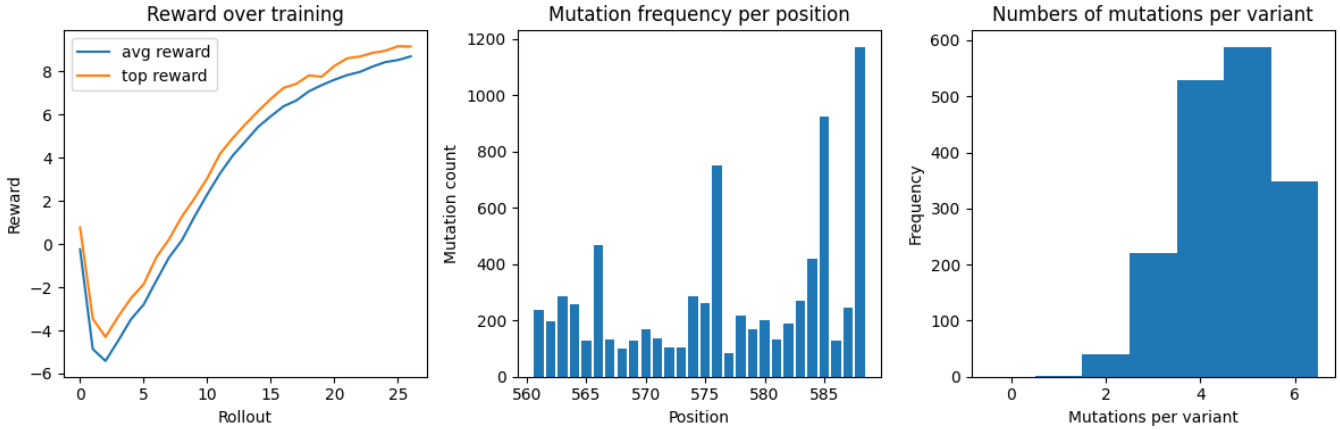

Figure XII: A2C TD(0.5) training metrics

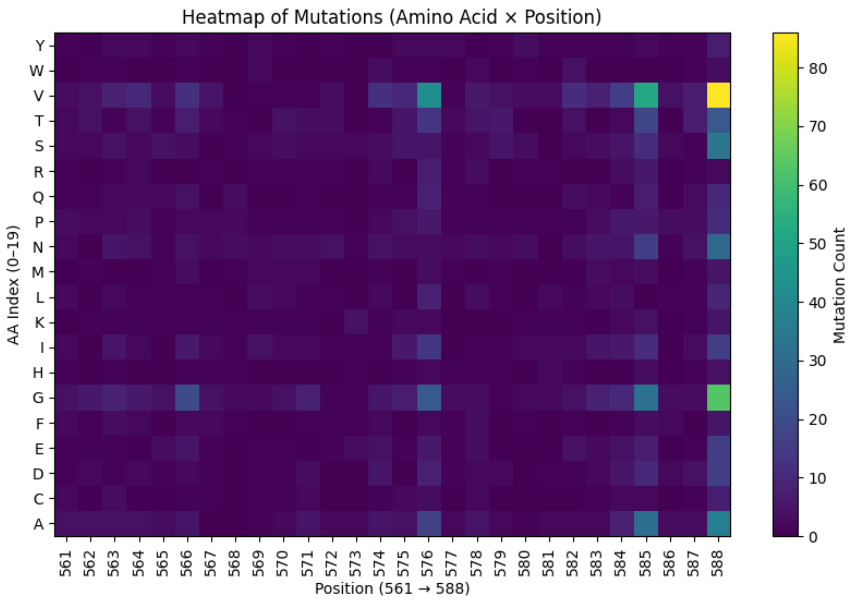

Figure XIII: A2C TD(0.5) heatmap of mutations over training

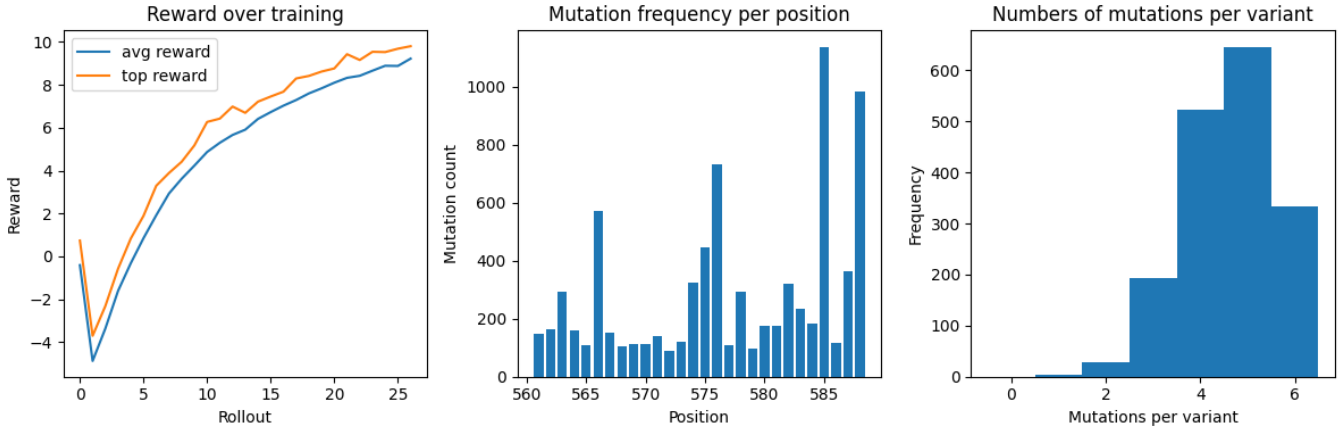

Figure XIV: A2C TD(0) training metrics

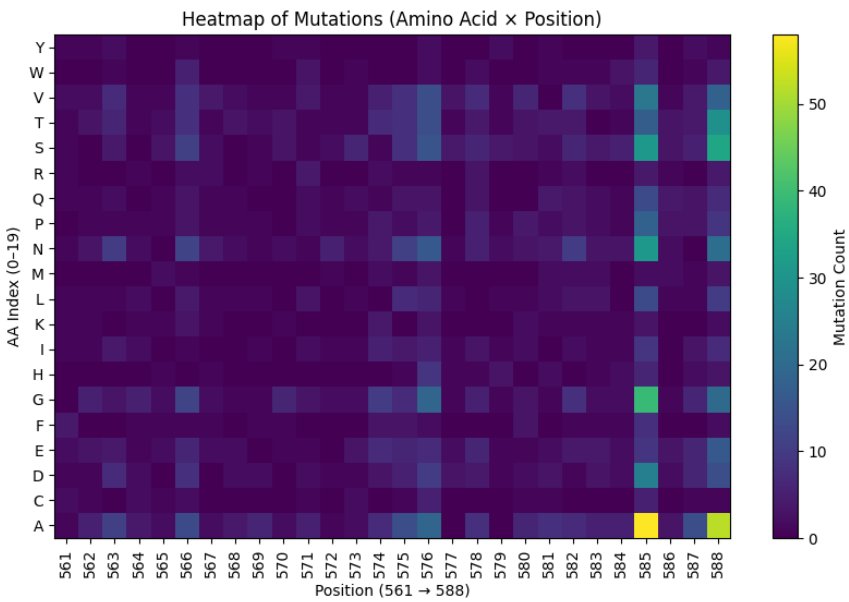

Figure XV: A2C TD(0) heatmap of mutations over training

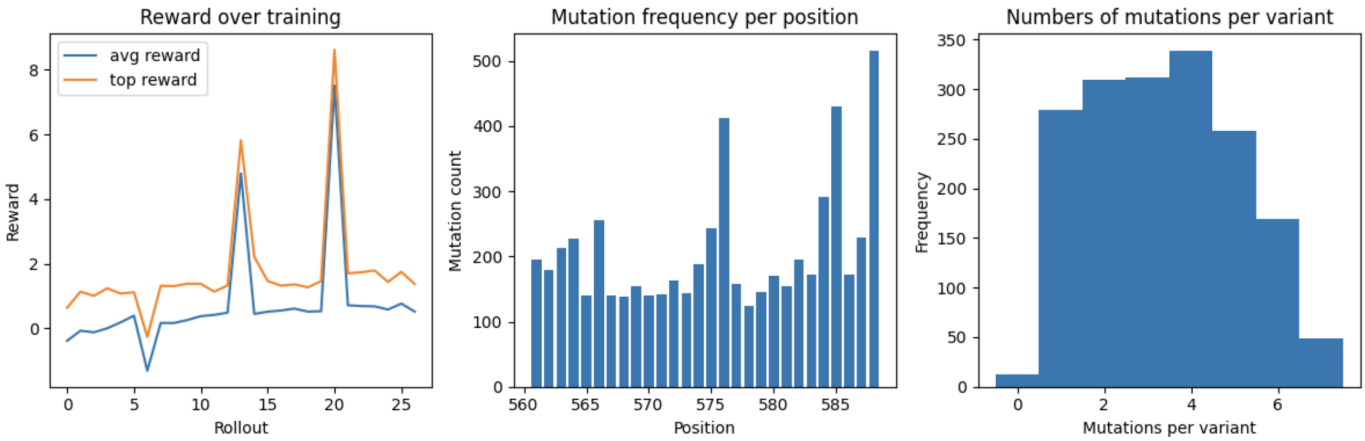

FIGURE XVI: PPO TRAINING METRICS WITH OUT-OF-SYNC CONDITION `STEP_COUNT` > 6

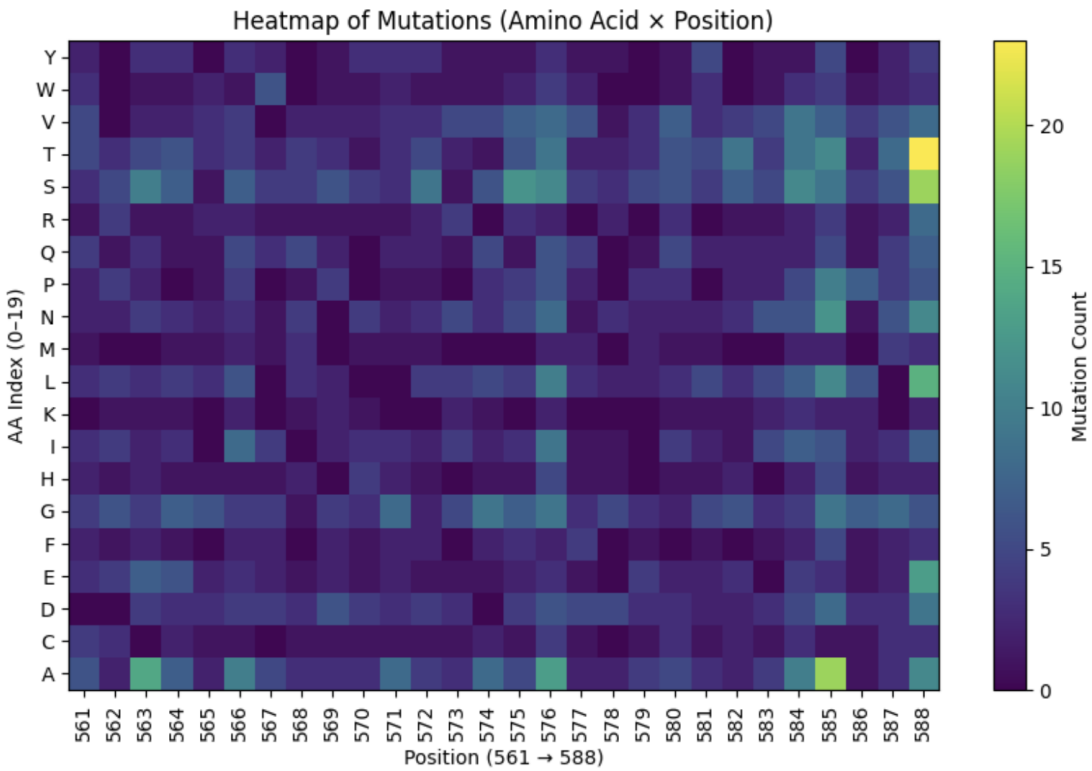

FIGURE XVII: PPO HEATMAP OF MUTATIONS OVER TRAINING WITH OUT-OF-SYNC CONDITION `STEP_COUNT` > 6

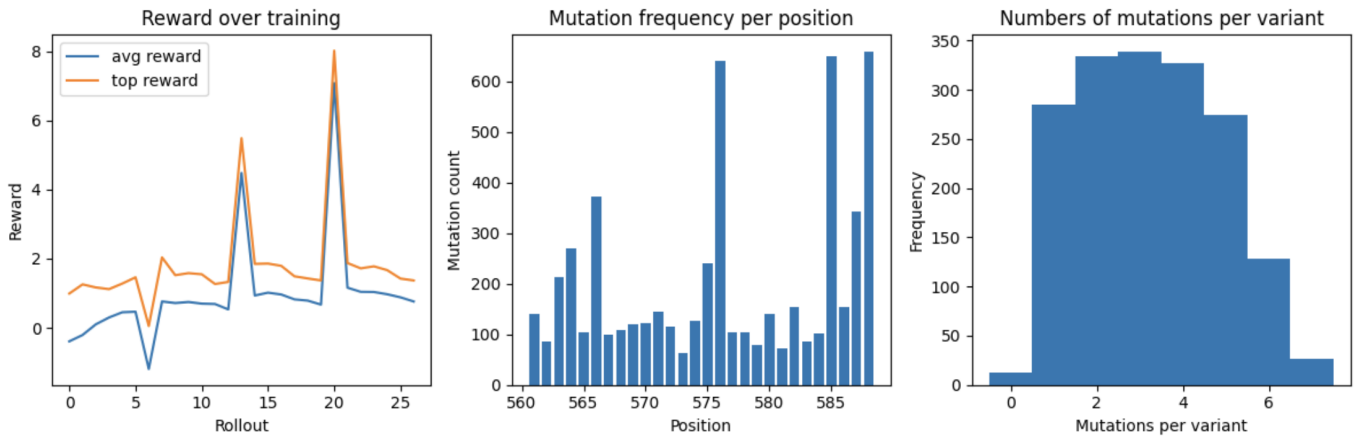

FIGURE XVIII: A2C TD(1) TRAINING METRICS WITH OUT-OF-SYNC CONDITION STEP_COUNT > 6

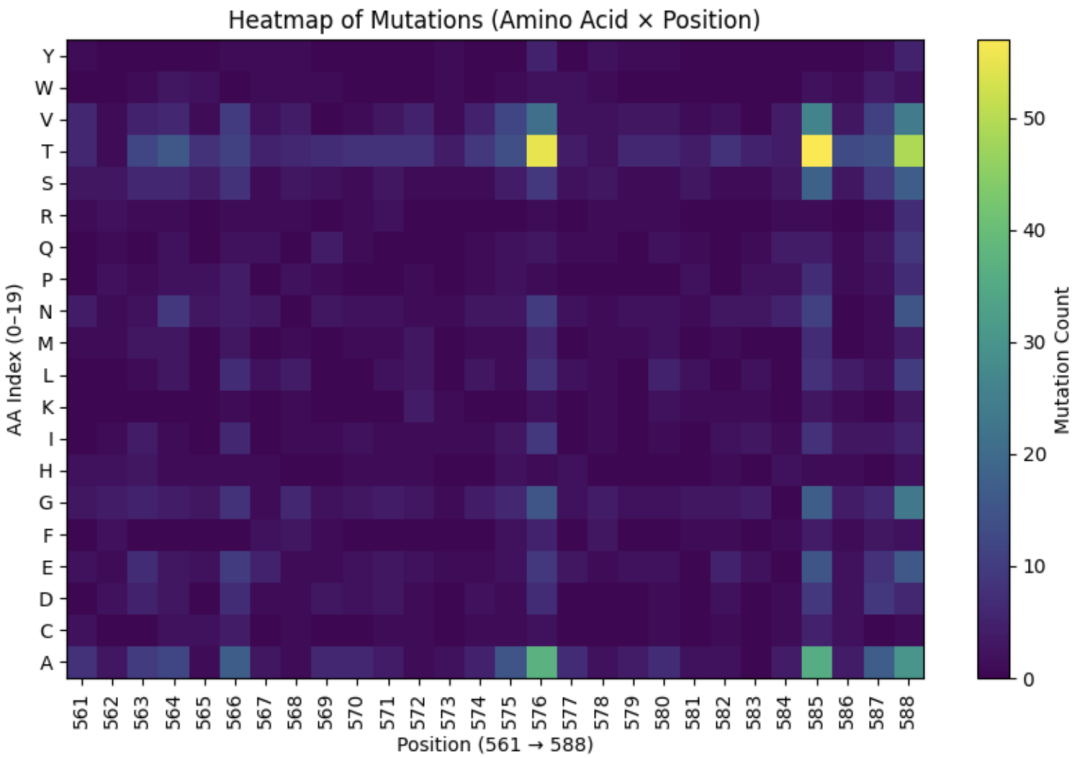

FIGURE XIX: A2C TD(1) HEATMAP OF MUTATIONS OVER TRAINING WITH OUT-OF-SYNC CONDITION STEP_COUNT > 6

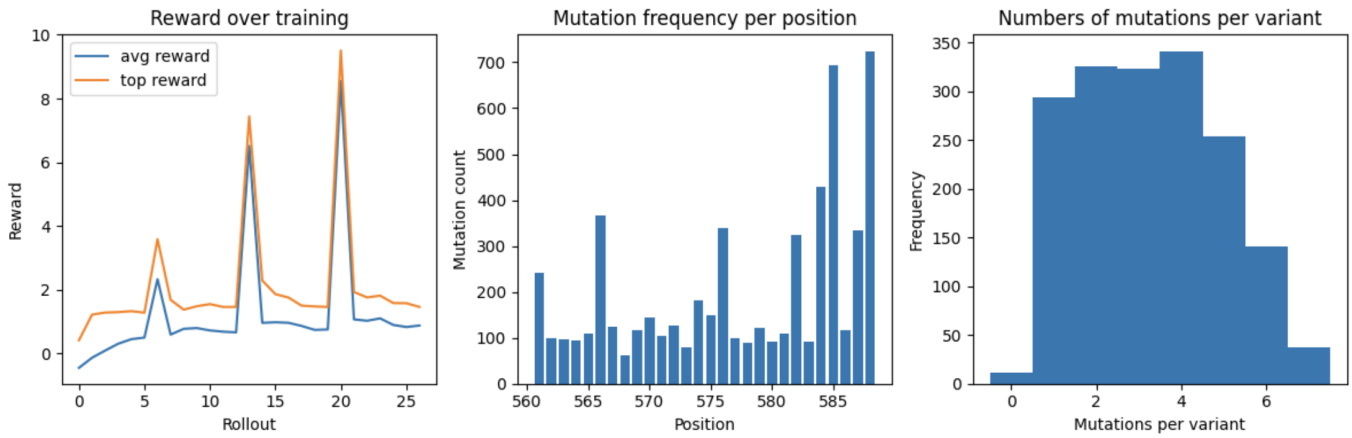

FIGURE XX: A2C TD(0.5) TRAINING METRICS WITH OUT-OF-SYNC CONDITION `STEP_COUNT` > 6

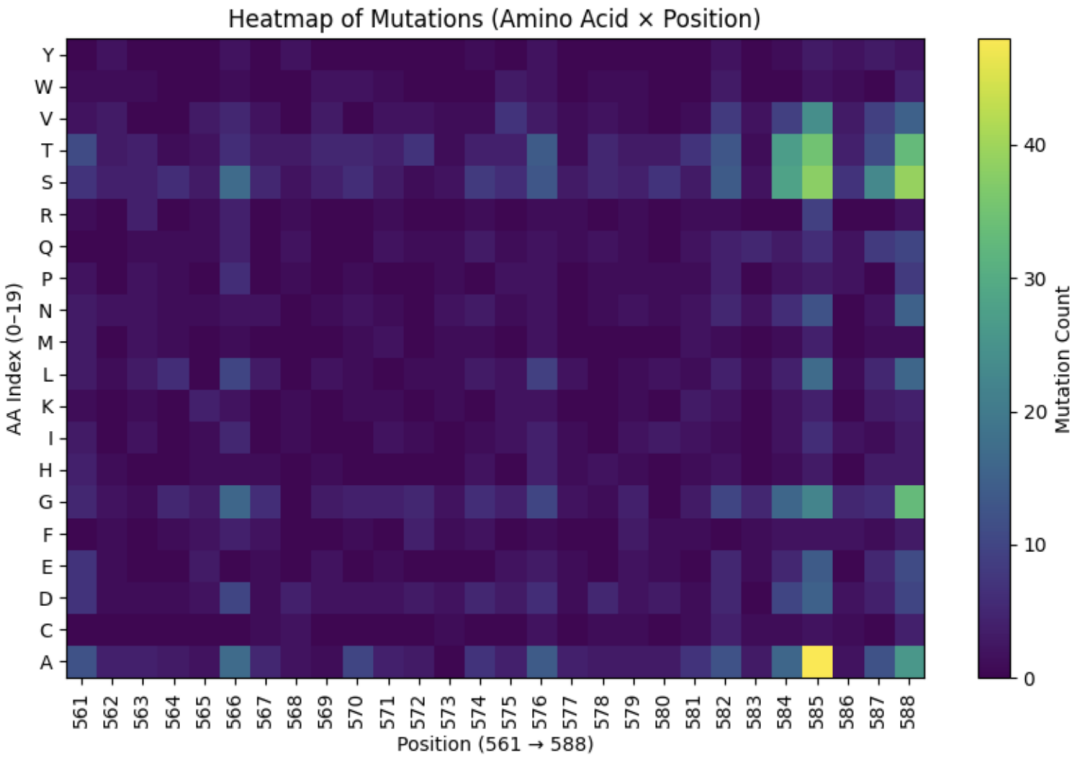

FIGURE XXI: A2C TD(0.5) HEATMAP OF MUTATIONS OVER TRAINING WITH OUT-OF-SYNC CONDITION `STEP_COUNT` > 6

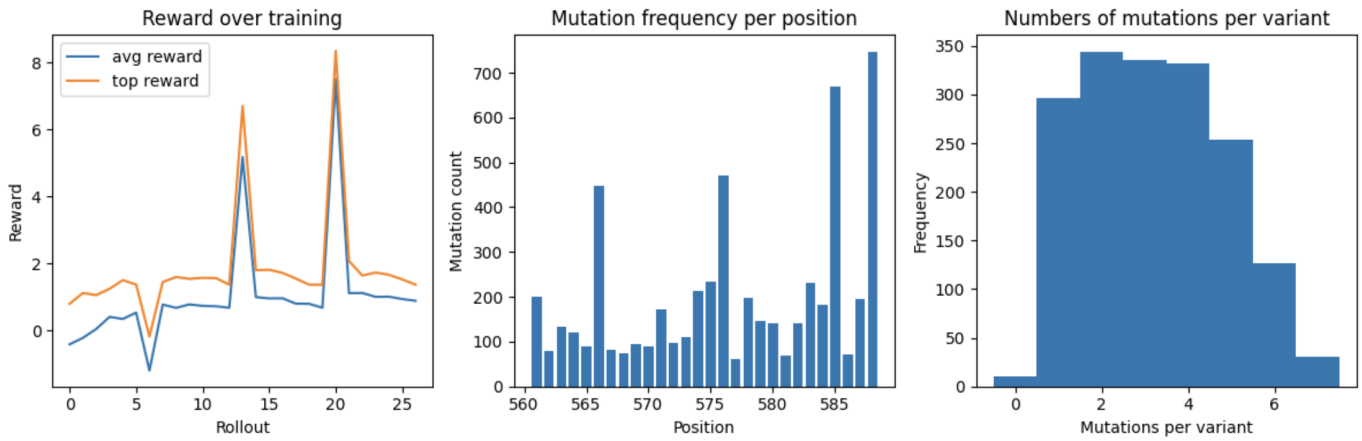

FIGURE XXII: A2C TD(0) TRAINING METRICS WITH OUT-OF-SYNC CONDITION STEP_COUNT > 6

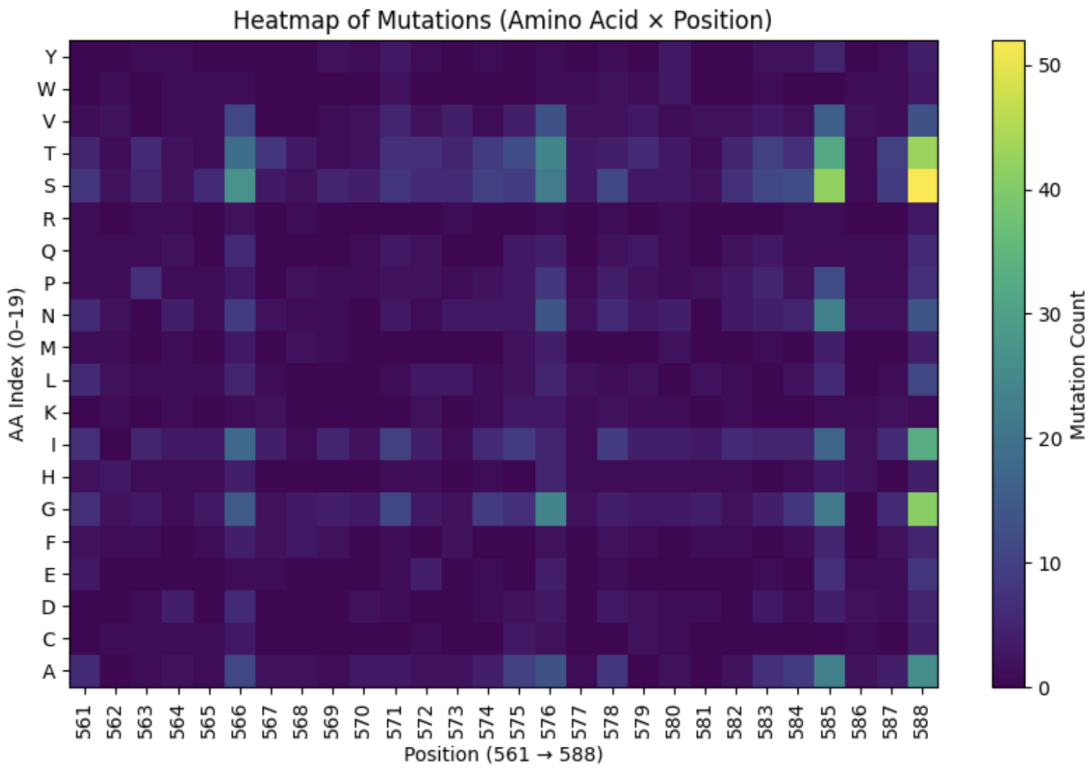

FIGURE XXIII: A2C TD(0) HEATMAP OF MUTATIONS OVER TRAINING WITH OUT-OF-SYNC CONDITION STEP_COUNT > 6

Best AAV2 variant sequences produced from each algorithm after sampling 10 variants from the learned policy:

A2C TD(0)
MAADGYLPDWLEDTLSEGIRQWWKLKPGPPPPKPAERHKDDSRGLVLPGYKYLGPFNGLDKGEPVNEADA
AALEHDKAYDRQLDSGDNPYLKYNHADAEFQERLKEDTSFGGNLGRAVFQAKKRVLEPLGLVEEPVKTAP
GKKRPVEHSPVEPDSSSGTGKAGQQPARKRLNFGQTGDADSVPDPQPLGQPPAAPSGLGTNTMATGSGAP
MADNNEGADGVGNSSGNWHCDSTWMGDRVITTSTRTWALPTYNNHLYKQISSQSGASNDNHYFGYSTPWG
YFDFNRFHCHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVKEVTQNDGTTTIANNLTSTVQVFTDSEYQLPY
VLGSAHQGCLPPFPADVFMVPQYGYLTLNNGSQAVGRSSFYCLEYFPSQMLRTGNNFTFSYTFEDVPFHSSY
AHSQSLDRLMNPLIDQYLYYLSRTNTPSGTTTQSRLQFSQAGASDIRDQSRNWLPGPCYRQQRVSKTSADNN
NSEYSWTGATKYHLNGRDSLVNPGPAMASHKDDEEKFFPQSGVLIFGKQGSEKTNVDIEKVMITDEEEIRTT
NIVATVAAGSVSTNLQTANRQAATADVNTQGVLPGMVWQDRDVYLQGPIWAKIPHTDGHFHPSPLMGGFG
LKHPPPQILIKNTPVPANPSTTFSAAKFASFITQYSTGQVSVEIEWELQKENSKRWNPEIQYTSNYNKSVNVDF
TVDTNGVYSEPRPIGTRYLTRNL
Reward: 1.789

A2C TD(0.5)
MAADGYLPDWLEDTLSEGIRQWWKLKPGPPPPKPAERHKDDSRGLVLPGYKYLGPFNGLDKGEPVNEADA
AALEHDKAYDRQLDSGDNPYLKYNHADAEFQERLKEDTSFGGNLGRAVFQAKKRVLEPLGLVEEPVKTAP
GKKRPVEHSPVEPDSSSGTGKAGQQPARKRLNFGQTGDADSVPDPQPLGQPPAAPSGLGTNTMATGSGAP
MADNNEGADGVGNSSGNWHCDSTWMGDRVITTSTRTWALPTYNNHLYKQISSQSGASNDNHYFGYSTPWG
YFDFNRFHCHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVKEVTQNDGTTTIANNLTSTVQVFTDSEYQLPY
VLGSAHQGCLPPFPADVFMVPQYGYLTLNNGSQAVGRSSFYCLEYFPSQMLRTGNNFTFSYTFEDVPFHSSY
AHSQSLDRLMNPLIDQYLYYLSRTNTPSGTTTQSRLQFSQAGASDIRDQSRNWLPGPCYRQQRVSKTSADNN
NSEYSWTGATKYHLNGRDSLVNPGPAMASHKDDEEKFFPQSGVLIFGKQGSEKTNVDIEKVMITDEEEIRTT
NPVATEVYGSVSTNLGTVNGQAATADVNTQGVLPGMVWQDRDVYLQGPIWAKIPHTDGHFHPSPLMGGFG
LKHPPPQILIKNTPVPANPSTTFSAAKFASFITQYSTGQVSVEIEWELQKENSKRWNPEIQYTSNYNKSVNVDF
TVDTNGVYSEPRPIGTRYLTRNL
Reward: 1.5822

A2C TD(1)
MAADGYLPDWLEDTLSEGIRQWWKLKPGPPPPKPAERHKDDSRGLVLPGYKYLGPFNGLDKGEPVNEADA
AALEHDKAYDRQLDSGDNPYLKYNHADAEFQERLKEDTSFGGNLGRAVFQAKKRVLEPLGLVEEPVKTAP
GKKRPVEHSPVEPDSSSGTGKAGQQPARKRLNFGQTGDADSVPDPQPLGQPPAAPSGLGTNTMATGSGAP
MADNNEGADGVGNSSGNWHCDSTWMGDRVITTSTRTWALPTYNNHLYKQISSQSGASNDNHYFGYSTPWG
YFDFNRFHCHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVKEVTQNDGTTTIANNLTSTVQVFTDSEYQLPY
VLGSAHQGCLPPFPADVFMVPQYGYLTLNNGSQAVGRSSFYCLEYFPSQMLRTGNNFTFSYTFEDVPFHSSY
AHSQSLDRLMNPLIDQYLYYLSRTNTPSGTTTQSRLQFSQAGASDIRDQSRNWLPGPCYRQQRVSKTSADNN
NSEYSWTGATKYHLNGRDSLVNPGPAMASHKDDEEKFFPQSGVLIFGKQGSEKTNVDIEKVMITDEESIRTT
NPGATEQTGSVSTNLQAGNRQAATADVNTQGVLPGMVWQDRDVYLQGPIWAKIPHTDGHFHPSPLMGGF
GLKHPPPQILIKNTPVPANPSTTFSAAKFASFITQYSTGQVSVEIEWELQKENSKRWNPEIQYTSNYNKSVNV
DFTVDTNGVYSEPRPIGTRYLTRNL
Reward: 1.325

PPO
MAADGYLPDWLEDTLSEGIRQWWKLKPGPPPPKPAERHKDDSRGLVLPGYKYLGPFNGLDKGEPVNEADA
AALEHDKAYDRQLDSGDNPYLKYNHADAEFQERLKEDTSFGGNLGRAVFQAKKRVLEPLGLVEEPVKTAP
GKKRPVEHSPVEPDSSSGTGKAGQQPARKRLNFGQTGDADSVPDPQPLGQPPAAPSGLGTNTMATGSGAP
MADNNEGADGVGNSSGNWHCDSTWMGDRVITTSTRTWALPTYNNHLYKQISSQSGASNDNHYFGYSTPWG
YFDFNRFHCHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVKEVTQNDGTTTIANNLTSTVQVFTDSEYQLPY
VLGSAHQGCLPPFPADVFMVPQYGYLTLNNGSQAVGRSSFYCLEYFPSQMLRTGNNFTFSYTFEDVPFHSSY
AHSQSLDRLMNPLIDQYLYYLSRTNTPSGTTTQSRLQFSQAGASDIRDQSRNWLPGPCYRQQRVSKTSADNN
NSEYSWTGATKYHLNGRDSLVNPGPAMASHKDDEEKFFPQSGVLIFGKQGSEKTNVDIEKVMITDEEEIRTS
NTVATEQTGSVSTNLQTGNTQAATADVNTQGVLPGMVWQDRDVYLQGPIWAKIPHTDGHFHPSPLMGGF
GLKHPPPQILIKNTPVPANPSTTFSAAKFASFITQYSTGQVSVEIEWELQKENSKRWNPEIQYTSNYNKSVNV
DFTVDTNGVYSEPRPIGTRYLTRNL
Reward: 2.169